# GIL NAVA ZARAGOZA

**Ford - Vehicle Hardware Engineering - Switches Applications Engineer | Automotive Systems Engineer | MSc Engineering Business Management**
**United States TN Visa: U6828565 (Expires in June 27, 2028)**

Portfolio: https://gilnaza92.github.io/
Email: gilnavazaragoza@gmail.com | Phone: +52 744 909 0832 | LinkedIn: linkedin.com/in/GilNavaZaragoza
Location: Mexico | Open to global opportunities

## SUMMARY

Automotive Electromechanical Systems Engineer with 7+ years specializing in interior mechanisms, switch applications, and premium user experience design. Expert in electromechanical integration combining mechanical, electrical, and software systems for luxury vehicle interiors. Proven expertise in kinematic analysis, infotainment validation, mechanism design , and cross-functional collaboration with design studios. Proficient in CATIA, 3D CAD modeling, GD&T, manufacturing processes, and LIN/CAN protocols. Strong background in developing high-quality interior products from concept through production.

## PROFESSIONAL EXPERIENCE

### Ford – Vehicle Hardware Engineering - Switches Applications Engineer (Nov 2023 – Present)

- Manage five vehicle line projects, resolving issues related to switch application systems across current and new models, drives on-time completion of tasks, consistently meeting vehicle program milestones.
- Develop and implement cost and complexity reduction initiatives for switches, increasing assembly line efficiency and saving over $2.5M per year during production phases.
- Generate and present ICA/PCA alternative matrix analyses to senior management, strategically mitigating body interior switch-related build stoppages.
- Lead the 25MY new model launch at the Ohio and Kentucky Truck Assembly Plants (8 months in Louisville, KY; 4 months in Cleveland, OH).
- Manage cross-functional body interior assessments (Appearance Approval, Blue Tag, and Studio reviews), ensuring switches perfect assemblies in cockpit / bezel & interior harmony.
- Spearheaded quality claims projects for switches, achieving a 77% reduction in cost per repair.
- Audit bill of materials (BOM) and manage change control to maintain product accuracy and compliance.

### Ford – Chassis - Steering Applications Engineer (Nov 2022 – Nov 2023)

- Collaborate with counterparts in Ford North America and Ford of Europe to validate, launch, and maintain Electric Power Assisted Steering (EPAS) systems, including steering column, intermediate shaft, floor seal, gear & linkage assembly, and heat shield.
- Monitor warranty performance of steering components, driving design improvements to resolve issues and prevent recurrence.
- Identify cost-saving opportunities within components and implement more efficient solutions in partnership with suppliers (Bosch, ZF, Thyssenkrupp).
- Coordinate bill of materials (BOM) and change control for delegated parts, ensuring accuracy and compliance.
- Analyze the impact of proposed design and production changes on system performance and compliance with engineering standards.
- Contribute to the successful implementation of EPAS running change screws in current vehicle production at the Hermosillo Stamping & Assembly Plant (4 months supporting new model launches).
- Develop innovation disclosures to enhance current product designs.

### Ford – Powertrain - Controls & Calibration Engineer (Jun 2021 – Nov 2022)

- Integrate sub-system calibrations for high-performance ICE and EV propulsion software, ensuring compliance with global emissions standards for Ford and Lincoln vehicles.
- Support and develop powertrain calibrations for ECU-controlled actuators and sensors (PCM, TCM, DLCM) using Candela Studio and CANape Vector tools.
- Optimize the calibration process by designing and implementing Python-based GUIs, reducing calibration time by 25%.
- Manage and maintain project tools to support collaborative data analysis initiatives.

### Stellantis – Mechatronics - Electronic Power Steering (EPS) Design & Release engineer (Oct 2020 – Jun 2021)

- Lead Electronic Power Steering (EPS) software readiness, coordinating with suppliers and product engineers to ensure quality in software releases, flashing processes, material traceability, updates, and calibrations.
- Develop, analyze, and validate DVP&R (Design Verification Plan & Report) in collaboration with steering suppliers (ZF, Nexteer).
- Support End-of-Line and Vehicle Dynamics teams with EPS module calibration.
- Provide chassis engineering support during prototype pilot launches at the Saltillo Van Assembly Plant, implementing ICA/PCA solutions for steering and chassis issues on RAM ProMaster ICE/EV (2023) and RAM 1500/2500 (2023).
- Audit and update Bill of Materials (BOM) to reflect engineering changes (DCR / TVM / SQ), ensuring correct part levels and usage.

### Ford – Infotainment and Connectivity DV Engineer (May 2018 – Oct 2020)

- Verify connectivity of multiple ECUs (TCU, IPC, SYNC, APIM, BCM) in alignment with Ford and Lincoln electronic architectures and global specifications.
- Identify software defects for the FordPass app through root cause analysis and defect classification.
- Validate, report, and track infotainment software bugs with suppliers using communication protocols (CAN, CAN FD, Ethernet, LIN, HS, MS).
- Assess software engineering changes resulting from issues found in prior authentication rounds.
- Multiple ECU´s connectivity verification, such as: TCU, IPC, SYNC, APIM,BCM, in agreement with Ford and Lincoln vehicle electronic architectures based on Ford global specifications.
- CAPL developed tools to fully automatize CANalyzer signals & messages databases to reduce the amount of time invested during testing rounds. From 30 to 2 hours. (93.33%)
- Support and mentor junior engineers through training and technical guidance.
- Features tested & validated: Wi-fi Hotspot, Fleet telematics , Power Management, Commands and controls, Control my car, Vehicle Health Alerts, Digital Owners Manuals, Electrical Vehicle features (On board testing), bluetooth, GPS, Online Traffic, audio, Car Play, Android Auto, Ford Pass App, OTA updates.

## EDUCATION & CERTIFICATIONS

**MSc Engineering Business Management** – Universidad Anáhuac México Norte (2023-2025)
**Automotive Systems Engineering** – Instituto Politécnico Nacional (2013-2017)
**Certifications:** Six Sigma Green Belt · HEV Electrified Powertrain · ISTQB · Python Programming · Agile & Scrum · Microsoft Office Advanced

## TECHNICAL SKILLS

**CAD/CAE:** CATIA V5, NX, SolidWorks, Abaqus, Adams, Ansys, 3DX | **Validation:** CANalyzer, Candela, Ford Diagnostic Eng. Tool, Ford HMI, Simulink, Matlab, Teamcenter, WERS
**Programming:** Python (automation/UI), MATLAB scripting, IntelliJ, GitHub, Copilot, Ford Telematics Protocol | **Methods:** Six Sigma, 8D/5D, Ishikawa, Agile, Scrum Master
**Languages:** Spanish (Native), English (Professional)

## CORE COMPETENCIES & EXPERTISE

**System Design:** Electromechanical System Design · Interior Mechanism Development · Kinematic Analysis · Cross-functional Collaboration
System Integration; Embedded Diagnostics; Validation Planning; Data Analysis; Calibration; Continuous Improvement; Launch & Software Readiness; Change Control; Root Cause; Supplier Coordination; Process Optimization.
**Leadership:** Communication · Teamwork · Creativity · Design Studio Coordination · Supplier Management · Manufacturing Process Optimization